

An Audit of *Graph Transfer Learning-Based Attack Detection in Cyber-Physical Water Distribution Systems*

ATK Evidence

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Abstract

This report audits the principal numerical and mechanistic claims of a graph-transformer attack detector for water distribution systems. It is not solely a reproduction attempt. Reproduction alone is a weak instrument: it invites the reply that the implementation was at fault, an objection that was correct at one point during this audit and is documented as such. The audit therefore proceeds in tiers, beginning with whether the reported numbers cohere internally, then whether the detection task is non-trivial at all, then whether each claimed component contributes, and only finally whether the headline pattern reproduces. Every experiment is designed so that its outcome is not controlled by the expected answer, and results that disconfirm the audit’s own expectations are reported as such. Conclusions are bounded to the artefact: this report makes no claim about author intent.

1 Scope and testable claims

The target paper reports F1, accuracy, and detection rate for nine detectors across three graph sizes (10, 20, 31 nodes) of the C-Town water network, and claims (i) a monotone improvement with graph size, (ii) a large margin for the proposed spatio-temporal graph transformer (STGT) over conventional baselines, (iii) a further gain from transfer learning across graph sizes, and (iv) a mechanism: that spatial and temporal dependencies are too difficult for a feed-forward network to capture, which is why a graph convolution and a transformer are required.

Claim (iv) is testable independently of whether any number reproduces, and is treated here as the primary target. Claims (i)–(iii) are evaluated against distributions across seeds rather than point estimates, because measured run-to-run spread for these architectures is large relative to the margins claimed.

2 Source protocol reconstruction

Benign readings are the canonical DeepH2O C-Town SCADA series (31 sensors: 7 tank levels, 11 pump flows, 1 valve flow, 12 junction pressures). Malicious samples are produced by modifying benign readings with three printed equations: replay ($X_t^m = X_{t-\Delta t}^b$), denial of service ($X_t^m = X_{t-1}^b$, sustained), and data manipulation ($X_t^m = X_t^b + \delta$, $-5 \leq \delta \leq 5$ in steps of 0.2). Detection compares a one-step forecast against the observed reading, scores the residual by squared Mahalanobis distance against a benign Gaussian, averages over consecutive batches of size S , and thresholds using the validation set.

The reduced 10- and 20-node graphs are transcribed verbatim from the paper’s Figure 1 rather than recomputed. This matters: the paper names betweenness centrality as the reduction criterion, but a betweenness ranking retains node J422 and drops PU8, whereas Figure 1(a) retains PU8. The printed figure governs. Transcribed node sets are nested and their induced edges match the drawing exactly (12, 24, and 44 edges).

3 Method

All experiments run on a single cluster environment with fixed library versions; each job records SHA-256 of code and data and runs a regression suite as a preflight. Zero-parameter reference detectors are evaluated through the *identical* scoring, threshold, and metric path as every trained model, so any difference is attributable to the model rather than to the evaluation. Comparisons between detectors are made at a matched false-alarm rate; an earlier draft of this audit compared detectors at different operating points and overstated the trained models’ advantage as a result.

Ambiguities in the printed method are enumerated and pre-registered before results are examined, covering training-set class balance, the units of δ , the detection batch size, the threshold rule, the transformer placement, and the population used for the Mahalanobis fit.

4 Results

4.1 Internal consistency of the reported table

On the paper’s own stated balanced split, accuracy satisfies $ACC = (DR + SP)/2$ with $SP \leq 100\%$, so $ACC \leq (DR + 100)/2$. Twenty-four of twenty-seven reported cells exceed this ceiling. Applying the stricter joint check, which uses all three reported metrics, all twenty-seven fail.

A stronger test asks whether each reported triple is attainable from *any* integer confusion matrix on a balanced test set. Scanning per-class sizes from 70 (the size implied by the paper’s own description of 1,400 hours split 80/10/10) to 700, no cell is attainable at any size. The checker was validated against fifteen constructed confusion matrices and accepted all fifteen.

The deviation is systematic rather than random. Of the twenty-five cells where the joint identity is evaluable, twenty-five report an accuracy above what their own detection rate and F1 imply, and none below, with a mean inflation of +30.6 points (maximum +54.7). A sign test against chance gives $p = 6 \times 10^{-8}$. Isolated transcription errors and rounding are therefore excluded, as these would scatter in both directions.

4.2 Does any measurement protocol reproduce the table?

Showing that numbers are unattainable under the protocol a paper *describes* is escapable: a defender may argue the description was inaccurate. The stronger test asks whether *any* protocol reproduces them. Formally, under H_0 : “the reported triple arose from a real confusion matrix under some protocol”, we evaluate $P(\text{table} \mid H_0)$ across a pre-registered space of 67,326 protocols per cell — every test-set size from 20 to 1,000, every class prevalence from 2% to 98%, and six metric-definition variants including balanced accuracy, detection-rate/precision swaps, negative-class and macro F1, and specificity reported in the detection-rate column.

Twenty of the twenty-seven cells are unreachable under *every* protocol searched. For those cells the likelihood under H_0 is exactly zero rather than small. Seven cells are individually reachable under some exotic setting, but the decisive constraint is joint: a paper has one protocol, not one

Attack class	Share	Best zero-parameter rule	Forecast-residual method
Manipulation	29%	100%	100%
Denial of service	29%	93% (AUC 0.987)	0%
Replay	43%	6%	0.6–6%

Table 1: Detection rate by attack class. The paper’s residual-based detector is structurally blind to denial of service: a forecaster supplied with frozen history predicts the frozen value, so the residual vanishes. A two-line stuck-sensor test separates the same attack at AUC 0.987.

per row. The best single protocol explains **one** cell of twenty-seven, and **no** protocol explains the table.

The searcher was validated by constructing confusion matrices at several sizes and prevalences, rounding their metrics, and confirming recovery in every case.

4.3 Is the task non-trivial?

Two zero-parameter detectors were evaluated: a per-sensor z -score on the observed reading, and persistence (predict the next reading as the last observed reading). Both achieve $F1 \approx 45$ at a false-alarm rate below 8%. Every baseline the paper reports — SVM (6.1), RF (17.5), LGBM (19.8), FFNN (20.9), AE (21.5), LSTM (26.0) at ten nodes — falls *below* a detector with no parameters.

Decomposing by attack class reveals why. Table 1 shows that the task is not one problem but three, with sharply different difficulty.

A composite of the two trivial rules (z -score or stuck-sensor) achieves $F1$ 71.5, accuracy 76.9, and detection rate 60.5 at a false-alarm rate of 7.9%, with zero parameters and no seed variance. Trained models reach $F1 \approx 70$ at a *higher* false-alarm rate when trained on benign windows only, and only 44–46 under the paper’s own stated prescription that every split carry equal samples per class. Under the paper’s literal training reading, every trained model scores below the zero-parameter composite.

4.4 Is replay detectable at all?

Replay reproduces genuine past readings, so a replayed window is itself a real benign window. Measuring nearest-neighbour distance from held-out test windows to a bank of attack-free training windows, replay is indistinguishable from normal data (AUC 0.455, median distance 9.75 versus 9.92 for normal), while denial of service (AUC 0.917) and manipulation (AUC 0.988) separate clearly. The replay ceiling is therefore a property of the data, not of any detector.

With replay undetectable, the achievable detection rate is bounded near $1 - (\text{replay share})$, which is 57.1% under the attack mix used here, against a reported 74.7% for STGT and 76.8% with transfer learning. The paper does not state its attack mix, so this bound is conditional on the mix rather than absolute.

4.5 Is the feed-forward network capacity-limited?

The paper attributes its architecture to the difficulty of spatio-temporal structure for a feed-forward network. Varying feed-forward capacity across a 272-fold parameter range (19,681 to 5,347,031 parameters) leaves performance flat: AUC 0.790, 0.785, 0.761, 0.787 and detection rate 56.7, 56.9, 41.9, 56.2 at a matched false-alarm rate. A two-layer network with fewer than twenty thousand parameters matches both the largest network and STGT.

Structure does exist in the data — mean cross-sensor correlation 0.18–0.24 and lag-one temporal correlation 0.65–0.74 on attack-free windows — so the premise that structure is present is not empty. What fails is the inference that capturing it requires the proposed architecture.

4.6 Where the paper’s premise holds

Reported because it disconfirms this audit’s expectation. Shuffling the C-Town adjacency, or removing its edges entirely, degrades performance by up to 18 F1 points: STGT at ten nodes scores 57.9 with the real topology, 51.9 shuffled, and 44.9 with no edges; TL-STGT at 31 nodes scores 63.6, 45.2, and 44.7 respectively. The graph is not decorative.

Transfer learning likewise shows a large gain at 31 nodes, +13.3 to +21.9 F1 over training from scratch. Two qualifications apply. The comparison is budget-confounded, since the transferred model is pretrained on the smaller graphs and sees roughly three times the total training. And under benign-only training the gain collapses to +3.1, indicating that transfer mainly buys robustness to the paper’s attack-corrupted training prescription rather than better detection.

4.7 Configuration dominates model choice

The paper states that every split carries an equal number of samples per class, so half of the training data is attack-corrupted. Training instead on benign windows changes F1 at 31 nodes by roughly 25 points: 44.1 versus 70.3 for the feed-forward network, 46.0 versus 71.2 for the recurrent baseline, 44.6 versus 70.8 for the graph baseline, 46.1 versus 67.8 for the proposed model, and 68.0 versus 70.9 with transfer learning. Every figure in this report therefore states its configuration; an unattributed value is meaningless at that sensitivity.

4.8 Stability

Across five seeds on identical hardware, software, and data, TGCN spans 23.7 F1 points and STGT 11.8, while feed-forward and recurrent baselines remain within 2–8. STGT’s denial-of-service detection alone swings from 5.8% to 95.8%. The paper reports single values to one decimal place with no dispersion.

5 Corrections to this audit

Two earlier conclusions of this audit were wrong and are retracted. First, a reported 46-point shortfall against the paper was substantially caused by a defect in our own threshold selection, which drew reference samples from validation windows contaminated by adjacent attacks and inflated the threshold by roughly four orders of magnitude; corrected, the shortfall is far smaller. Second, an identical-metrics collapse across architectures was attributed to the scale incoherence of δ ; the corrected runs show that the threshold defect alone produced it. Third, an earlier draft quoted benign-training results without stating the configuration, while the paper’s literal reading trains on the balanced split; the two differ by roughly 25 F1. Fourth, this audit expected the graph topology to carry no signal, and it does. All are recorded in the evidence log rather than silently amended.

6 Limitations

The benign series is the canonical DeepH2O C-Town series rather than the paper’s own unpublished 1,400-hour run. Attack duration, the replay offset Δt , the sliding-window length, and the detection

rule for the shallow classifiers are not specified by the paper and are documented assumptions. The attack mix is ours, which conditions the detection-rate ceiling. The interpretation sweep over pre-registered ambiguity axes is incomplete.

7 Verdict

Bounded to the artefact and to the finite space tested: the reported table is not reproducible by any correct implementation. No confusion matrix on the described test set yields the printed numbers; the deviation from internal consistency is systematic rather than accidental; and an exhaustive search over 67,326 measurement protocols per cell finds no single protocol that reproduces the table, with twenty of twenty-seven cells unreachable under every protocol searched. Independently of the numbers, the claimed mechanism does not hold: the detection task decomposes into one class requiring no model, one class that the paper’s own detector cannot express but a two-line rule solves, and one class that is undetectable in principle. A composite of two zero-parameter rules matches the proposed architecture at a lower false-alarm rate and with no seed variance.

This report does not assert how the reported numbers arose. Establishing that no measurement process in the searched space produces them does not distinguish deliberate fabrication from catastrophic process failure — placeholder values never replaced, or a table carried over from a draft — since both yield numbers that were never measured. That distinction requires evidence outside the paper, which this report does not have and does not claim.

8 Reproducibility materials

Code, pre-registered ambiguity register, evidence log with retractions, and job records with code and data checksums accompany this report.